4.2.6 Solved Problems:
Special Continuous Distributions

Problem 1

Suppose the number of customers arriving at a store obeys a Poisson distribution with an average of λ customers per unit time. That is, if Y is the number of customers arriving in an interval of length t, then $Y \sim Poisson(\lambda t)$. Suppose that the store opens at time $t = 0$. Let X be the arrival time of the first customer. Show that $X \sim Exponential(\lambda)$.

Solution

- We first find $P(X > t)$:

$$\begin{aligned} P(X > t) &= P(\text{No arrival in } [0, t]) \\ &= e^{-\lambda t} \frac{(\lambda t)^0}{0!} \\ &= e^{-\lambda t}. \end{aligned}$$

Thus, the CDF of X for $x > 0$ is given by

$$F_X(x) = 1 - P(X > x) = 1 - e^{-\lambda x},$$

which is the CDF of $Exponential(\lambda)$. Note that by the same argument, the time between the first and second customer also has $Exponential(\lambda)$ distribution. In general, the time between the k'th and $k+1$'th customer is $Exponential(\lambda)$.

Problem 2 (Exponential as the limit of Geometric)

Let $Y \sim Geometric(p)$, where $p = \lambda \Delta$. Define $X = Y\Delta$, where $\lambda, \Delta > 0$. Prove that for any $x \in (0, \infty)$, we have

$$\lim_{\Delta \to 0} F_X(x) = 1 - e^{-\lambda x}.$$

Solution

- If $Y \sim Geometric(p)$ and $q = 1 - p$, then

$$\begin{aligned} P(Y \leq n) &= \textstyle\sum_{k=1}^{n} pq^{k-1} \\ &= p.\, \tfrac{1-q^n}{1-q} = 1 - (1-p)^n. \end{aligned}$$

Then for any $y \in (0, \infty)$, we can write

$$P(Y \leq y) = 1 - (1-p)^{\lfloor y \rfloor},$$

where $\lfloor y \rfloor$ is the largest integer less than or equal to y. Now, since $X = Y\Delta$, we have

$$\begin{aligned} F_X(x) &= P(X \leq x) \\ &= P\left(Y \leq \tfrac{x}{\Delta}\right) \\ &= 1 - (1-p)^{\lfloor \frac{x}{\Delta} \rfloor} = 1 - (1 - \lambda\Delta)^{\lfloor \frac{x}{\Delta} \rfloor}. \end{aligned}$$

Now, we have

$$\begin{aligned}\lim_{\Delta\to 0} F_X(x) &= \lim_{\Delta\to 0} 1-(1-\lambda\Delta)^{\lfloor \frac{x}{\Delta}\rfloor} \\ &= 1-\lim_{\Delta\to 0}(1-\lambda\Delta)^{\lfloor \frac{x}{\Delta}\rfloor} \\ &= 1-e^{-\lambda x}.\end{aligned}$$

The last equality holds because $\frac{x}{\Delta}-1 \leq \lfloor \frac{x}{\Delta}\rfloor \leq \frac{x}{\Delta}$, and we know

$$\lim_{\Delta\to 0^+}(1-\lambda\Delta)^{\frac{1}{\Delta}} = e^{-\lambda}.$$

Problem 3

Let $U\sim Uniform(0,1)$ and $X=-\ln(1-U)$. Show that $X\sim Exponential(1)$.

Solution

- First note that since $R_U=(0,1)$, $R_X=(0,\infty)$. We will find the CDF of X. For $x\in(0,\infty)$, we have

$$\begin{aligned}F_X(x) &= P(X\leq x) \\ &= P(-\ln(1-U)\leq x) \\ &= P\left(\tfrac{1}{1-U}\leq e^x\right) \\ &= P(U\leq 1-e^{-x}) = 1-e^{-x},\end{aligned}$$

which is the CDF of an $Exponential(1)$ random variable.

Problem 4

Let $X\sim N(2,4)$ and $Y=3-2X$.

a. Find $P(X>1)$.
b. Find $P(-2<Y<1)$.
c. Find $P(X>2|Y<1)$.

Solution

- a. Find $P(X>1)$: We have $\mu_X=2$ and $\sigma_X=2$. Thus,

$$\begin{aligned}P(X>1) &= 1-\Phi\left(\tfrac{1-2}{2}\right) \\ &= 1-\Phi(-0.5)=\Phi(0.5)=0.6915\end{aligned}$$

  b. Find $P(-2<Y<1)$: Since $Y=3-2X$, using Theorem 4.3, we have $Y\sim N(-1,16)$. Therefore,

$$\begin{aligned}P(-2<Y<1) &= \Phi\left(\tfrac{1-(-1)}{4}\right)-\Phi\left(\tfrac{(-2)-(-1)}{4}\right) \\ &= \Phi(0.5)-\Phi(-0.25)=0.29\end{aligned}$$

  c. Find $P(X>2|Y<1)$:

$$\begin{aligned}P(X>2|Y<1) &= P(X>2|3-2X<1) \\ &= P(X>2|X>1)\end{aligned}$$

$$
\begin{aligned}
&= \frac{P(X>2, X>1)}{P(X>1)} \\
&= \frac{P(X>2)}{P(X>1)} \\
&= \frac{1-\Phi(\frac{2-2}{2})}{1-\Phi(\frac{1-2}{2})} \\
&= \frac{1-\Phi(0)}{1-\Phi(-0.5)} \\
&\approx 0.72
\end{aligned}
$$

Problem 5

Let $X \sim N(0, \sigma^2)$. Find $E|X|$

Solution

- We can write $X = \sigma Z$, where $Z \sim N(0,1)$. Thus, $E|X| = \sigma E|Z|$. We have

$$
\begin{aligned}
E|Z| &= \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} |t| e^{-\frac{t^2}{2}} \, dt \\
&= \frac{2}{\sqrt{2\pi}} \int_{0}^{\infty} |t| e^{-\frac{t^2}{2}} \, dt \qquad \text{(integral of an even function)} \\
&= \sqrt{\frac{2}{\pi}} \int_{0}^{\infty} t e^{-\frac{t^2}{2}} \, dt \\
&= \sqrt{\frac{2}{\pi}} \left[-e^{-\frac{t^2}{2}} \right]_0^{\infty} = \sqrt{\frac{2}{\pi}}
\end{aligned}
$$

Thus, we conclude $E|X| = \sigma E|Z| = \sigma \sqrt{\frac{2}{\pi}}$.

Problem 6

Show that the constant in the normal distribution must be $\frac{1}{\sqrt{2\pi}}$. That is, show that

$$
I = \int_{-\infty}^{\infty} e^{-\frac{x^2}{2}} \, dx = \sqrt{2\pi}.
$$

Hint: Write I^2 as a double integral in polar coordinates.

Solution

- Let $I = \int_{-\infty}^{\infty} e^{-\frac{x^2}{2}} \, dx$. We show that $I^2 = 2\pi$. To see this, note

$$
\begin{aligned}
I^2 &= \int_{-\infty}^{\infty} e^{-\frac{x^2}{2}} \, dx \int_{-\infty}^{\infty} e^{-\frac{y^2}{2}} \, dy \\
&= \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} e^{-\frac{x^2+y^2}{2}} \, dx dy.
\end{aligned}
$$

To evaluate this double integral we can switch to polar coordinates. This can be done by change of variables $x = r\cos\theta, y = r\sin\theta$, and $dxdy = rdrd\theta$. In particular, we have

$$
I^2 = \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} e^{-\frac{x^2+y^2}{2}} \, dx dy
$$

$$
\begin{aligned}
&= \int_0^\infty \int_0^{2\pi} e^{-\frac{r^2}{2}}\, r d\theta dr \\
&= 2\pi \int_0^\infty r e^{-\frac{r^2}{2}}\, dr \\
&= 2\pi \Big[-e^{-\frac{r^2}{2}} \Big]_0^\infty = 2\pi.
\end{aligned}
$$

Problem 7

Let $Z \sim N(0,1)$. Prove for all $x \geq 0$,

$$
\frac{1}{\sqrt{2\pi}} \frac{x}{x^2+1} e^{-\frac{x^2}{2}} \leq P(Z \geq x) \leq \frac{1}{\sqrt{2\pi}} \frac{1}{x} e^{-\frac{x^2}{2}}.
$$

Solution

- To show the upper bound, we can write

$$
\begin{aligned}
P(Z \geq x) &= \tfrac{1}{\sqrt{2\pi}} \int_x^\infty e^{-\frac{u^2}{2}}\, du \\
&\leq \tfrac{1}{\sqrt{2\pi}} \int_x^\infty \tfrac{u}{x} e^{-\frac{u^2}{2}}\, du \quad (\text{since } u \geq x > 0) \\
&= \tfrac{1}{\sqrt{2\pi}} \tfrac{1}{x} \Big[-e^{-\frac{u^2}{2}} \Big]_x^\infty \\
&= \tfrac{1}{\sqrt{2\pi}} \tfrac{1}{x} e^{-\frac{x^2}{2}}.
\end{aligned}
$$

To show the lower bound, let $Q(x) = P(Z \geq x)$, and

$$
h(x) = Q(x) - \frac{1}{\sqrt{2\pi}} \frac{x}{x^2+1} e^{-\frac{x^2}{2}}, \quad \text{for all } x \geq 0.
$$

It suffices to show that

$$
h(x) \geq 0, \quad \text{for all } x \geq 0.
$$

To see this, note that the function h has the following properties

1. $h(0) = \frac{1}{2}$;
2. $\lim\limits_{x \to \infty} h(x) = 0$;
3. $h'(x) = -\frac{2}{\sqrt{2\pi}} \left(\frac{e^{-\frac{x^2}{2}}}{(x^2+1)^2} \right) < 0$, for all $x \geq 0$.

Therefore, $h(x)$ is a strictly decreasing function that starts at $h(0) = \frac{1}{2}$ and decreases as x increases. It approaches 0 as x goes to infinity. We conclude that $h(x) \geq 0$, for all $x \geq 0$.

Problem 8

Let $X \sim Gamma(\alpha, \lambda)$, where $\alpha, \lambda > 0$. Find EX, and $Var(X)$.

Solution

- To find EX we can write

$$
\begin{aligned}
EX &= \int_0^\infty x f_X(x) dx \\
&= \int_0^\infty x \cdot \frac{\lambda^\alpha}{\Gamma\alpha} x^{\alpha-1} e^{-\lambda x} \mathrm{d}x \\
&= \frac{\lambda^\alpha}{\Gamma(\alpha)} \int_0^\infty x \cdot x^{\alpha-1} e^{-\lambda x} \mathrm{d}x \\
&= \frac{\lambda^\alpha}{\Gamma(\alpha)} \int_0^\infty x^{\alpha} e^{-\lambda x} \mathrm{d}x \\
&= \frac{\lambda^\alpha}{\Gamma(\alpha)} \frac{\Gamma(\alpha+1)}{\lambda^{\alpha+1}} && \text{(using Property 2 of the gamma function)} \\
&= \frac{\alpha\Gamma(\alpha)}{\lambda\Gamma(\alpha)} && \text{(using Property 3 of the gamma function)} \\
&= \frac{\alpha}{\lambda}.
\end{aligned}
$$

Similarly, we can find EX^2:

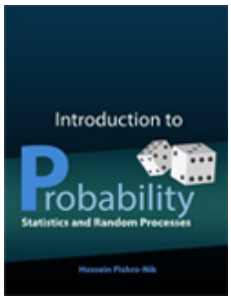
$$
\begin{aligned}
EX^2 &= \int_0^\infty x^2 \mathrm{d}x \\
&= \int_0^\infty x^2 \cdot \frac{\lambda^\alpha}{\Gamma(\alpha)} x^{\alpha-1} e^{-\lambda x} \mathrm{d}x \\
&= \frac{\lambda^\alpha}{\Gamma(\alpha)} \int_0^\infty x^2 \cdot x^{\alpha-1} e^{-\lambda x} \mathrm{d}x \\
&= \frac{\lambda^\alpha}{\Gamma(\alpha)} \int_0^\infty x^{\alpha+1} e^{-\lambda x} \mathrm{d}x \\
&= \frac{\lambda^\alpha}{\Gamma(\alpha)} \frac{\Gamma(\alpha+2)}{\lambda^{\alpha+2}} && \text{(using Property 2 of the gamma function)} \\
&= \frac{(\alpha+1)\Gamma(\alpha+1)}{\lambda^2\Gamma(\alpha)} && \text{(using Property 3 of the gamma function)} \\
&= \frac{(\alpha+1)\alpha\Gamma(\alpha)}{\lambda^2\Gamma(\alpha)} && \text{(using Property 3 of the gamma function)} \\
&= \frac{\alpha(\alpha+1)}{\lambda^2}.
\end{aligned}
$$

So, we conclude

$$
\begin{aligned}
Var(X) &= EX^2 - (EX)^2 \\
&= \frac{\alpha(\alpha+1)}{\lambda^2} - \frac{\alpha^2}{\lambda^2} \\
&= \frac{\alpha}{\lambda^2}.
\end{aligned}
$$

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The print version of the book is available on Amazon.



Practical uncertainty: *Userful Ideas in Decision-Making, Risk, Randomness, & AI*

